

Pressure in Solids and Liquids — Paper C

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formulae and no conversion factors are given on this paper.
- Several questions describe situations you may not have seen before — apply the ideas you already have.
- Take $g = 10 \text{ N/kg}$ throughout, and give pressures in pascals unless told otherwise.

SECTION 1 – THE SAME WEIGHT, A DIFFERENT FACE

1. A rectangular concrete block has a weight of 2400 N.

- (a)** It rests on a face measuring 1.5 m by 0.8 m. Calculate the pressure on the ground. [2]

- (b)** It is stood on a face measuring 0.8 m by 0.5 m instead. Calculate the new pressure. [2]

- (c)** State the factor by which the pressure changed, and say why. [1]

2. A kitchen knife is pressed onto a vegetable with a force of 45 N. The sharpened edge touches an area of 0.15 cm^2 .

(a) Convert the contact area into square metres. [2]

(b) Calculate the pressure under the edge. [2]

(c) The knife goes blunt and now touches 1.5 cm^2 . State the new pressure. [1]

SECTION 2 – UNDER THE SEA

3. A submarine has a window of area 0.05 m^2 . It dives to a depth of 60 m in seawater of density 1030 kg/m^3 . Atmospheric pressure is 100 000 Pa.

(a) Calculate the pressure due to the seawater alone. [2]

(b) Calculate the total pressure on the outside of the window. [2]

(c) Calculate the force the sea exerts on the window. [2]

4. Point X is 4 m below the surface of a freshwater lake and point Y is 12 m below it. The density of the water is 1000 kg/m^3 .

(a) Calculate the pressure due to the water at X and at Y. [2]

(b) Calculate the difference in pressure between the two points. [1]

(c) The lake is drained and replaced by a swimming pool with the same two depths but one tenth of the surface area. State and explain what happens to your answers. [2]

SECTION 3 – MACHINES THAT MULTIPLY FORCE

5. A hydraulic jack has a small piston of area 0.0004 m^2 and a large piston of area 0.03 m^2 . A force of 20 N is applied to the small piston.

(a) Calculate the pressure produced in the oil. [2]

(b) Calculate the force produced at the large piston. [2]

(c) State the factor by which the force has been multiplied. [1]

6. Explain why the fluid inside a hydraulic system must be a liquid and not a gas. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.